d) They undergo mitotic cell division.

e) They do not store food materials.

A. Types of Meristems based on position

On the basis of their position in the plant, meristems are of three types: Apical meristem, Intercalary meristem and Lateral meristem.

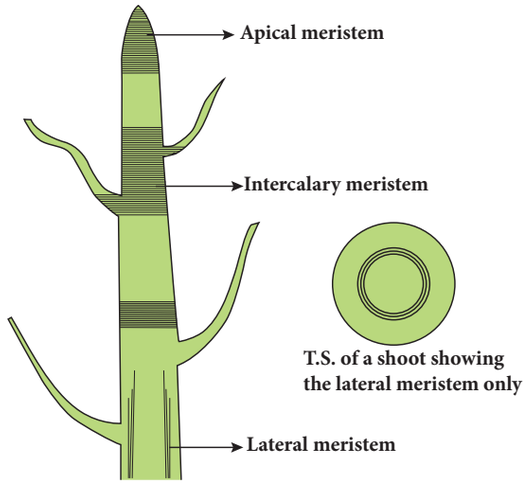


Figure 18.1 Longitudinal section of shoot apex showing location of meristems and young leaves.

Apical meristem: These are found at the apices or growing points of root and shoot and bring about increase in length.

Intercalary meristem: It lies between the region of permanent tissues and is part of primary meristem. It is found either at the base of leaf (e.g. pinus) or at the base of internodes (e.g. grasses).

Lateral Meristem: These are arranged parallel and causes the thickness of the plant part.

B. Functions of meristems

Meristems are actively dividing tissues of the plant, that are responsible for primary (elongation) and secondary (thickness) growth of the plant.

18.1.2 Permanent Tissues

Permanent tissues are those in which, growth has stopped either completely or for the time being. At times, they become meristematic partially or wholly. Permanent tissues are of two types, namely: simple tissue and complex tissue.

A. Simple Tissues

Simple tissues are homogeneous tissues composed of structurally and functionally similar cells. eg., Parenchyma, Collenchyma and Sclerenchyma.

Parenchyma: Parenchyma are simple permanent tissues composed of living cells. Parenchyma cells are thin walled, oval, rounded or polygonal in shape with well developed spaces among them. In aquatic plants, parenchyma possesses intercellular air spaces, and is named as Aerenchyma. When exposed to light, parenchyma cells may develop chloroplasts and are known as Chlorenchyma.

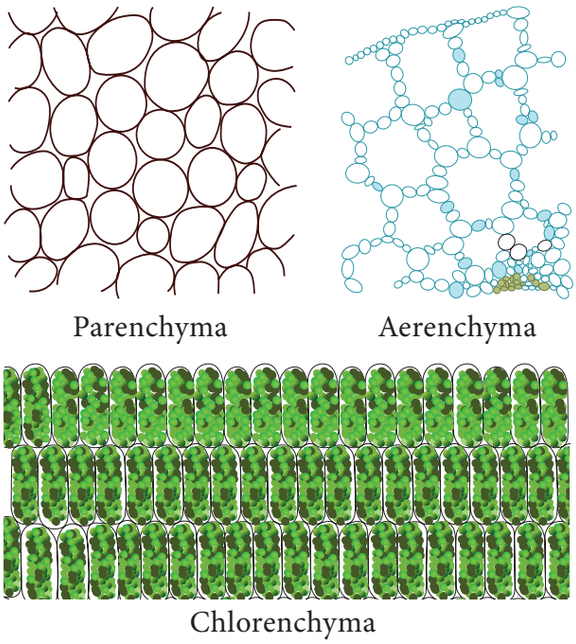


Figure 18.2 Types of Parenchyma

Parenchyma may store water in many succulent and xerophytic plants. It also serves the functions of storage of food reserves, absorption, buoyancy, secretion etc.,

DO YOU KNOW?

In potato, parenchyma vacuoles are filled with starch. In apple, parenchyma stores sugar

Collenchyma: Collenchyma is a living tissue found beneath the epidermis. Cells are elongated with unevenly thickened walls.

IX_SCI_EM_Unit-18_BIO.indd 211 30/11/2022 20:07:42